OLLSCOIL NA GAILLIMHE
UNIVERSITY OF GALWAY

Semester I Examinations 2023/24

Exam Code(s)	4BMS2, 4BS2, 1HA1, 1AL1
Exam(s)	4th Science
Module(s)	Algebraic Foundations of Quantum Computing
Module Code(s)	MA4102
Paper No.	1
Repeat Paper	No
External Examiner(s)	Prof. Colva Roney-Dougal
Internal Examiner(s)	Dr. Tobias Rossmann *Dr. Michael Mc Gettrick
Instructions	Answer **FOUR** questions.
Duration	2 hours
No. of Pages	3 pages (including this cover page)
Discipline	Mathematics

Requirements:

Release to Library	Yes ☑	No ☐
MCQ	Yes ☐	No ☑
Statistical/Log Tables	Yes ☑	No ☐

1. (i) Consider the **qutrit** state given by

$$|\psi\rangle = \frac{i}{3}\left(\sqrt{-7}\,|0\rangle + i\,|1\rangle + |2\rangle\right).$$

(a) What is the probability of measuring the particle in the $|1\rangle$ state?

(b) Calculate the density matrix ρ_ψ.

(c) If we operate on $|\psi\rangle$ using the unitary

$$U = \frac{1}{\sqrt{3}}\begin{pmatrix} 1 & 1 & 1 \\ 1 & \omega & \omega^2 \\ 1 & \omega^2 & \omega \end{pmatrix},$$

where $\omega = e^{2\pi i/3}$, calculate the subsequent probability of measuring $|1\rangle$.

[14 marks]

(ii) Consider the **qubit** state

$$|\psi\rangle = \frac{1}{\sqrt{5}}\left(i\,|0\rangle - 2\,|1\rangle\right).$$

(a) If $|\psi\rangle$ is represented on the Bloch Sphere, calculate the corresponding angles θ and ϕ.

(b) If we measure using the orthonormal basis $|+\rangle$ and $|-\rangle$, calculate the probability of obtaining $|+\rangle$.

[11 marks]

2. Suppose we write a bipartite (2-qubit) state as a column vector with 4 components, using the standard basis $|00\rangle = (1,0,0,0)^T$, etc. Calculate (a) the purity and (b) the entanglement (using the Von Neumann entropy of the reduced density matrix) for the second qubit in each of the following cases.

$$\text{(i)}\ \frac{1}{\sqrt{2}}\begin{pmatrix} -1 \\ 0 \\ 1 \\ 0 \end{pmatrix} \qquad \text{(ii)}\ \frac{1}{\sqrt{3}}\begin{pmatrix} -i \\ -1 \\ 0 \\ -1 \end{pmatrix} \qquad \text{(iii)}\ \frac{1}{\sqrt{2}}\begin{pmatrix} 0 \\ -1 \\ i \\ 0 \end{pmatrix} \qquad \text{(iv)}\ \frac{1}{\sqrt{7}}\begin{pmatrix} i \\ 2i \\ 1 \\ 1 \end{pmatrix}$$

[25 marks]

3. (i) A Phase Damping Channel is given by the Completely Positive Trace Preserving (CPTP) map

$$U(|10\rangle) = |10\rangle \qquad U(|11\rangle) = |11\rangle$$
$$U(|00\rangle) = \sqrt{1-p}\,|00\rangle + i\sqrt{p}\,|01\rangle$$
$$U(|01\rangle) = i\sqrt{p}\,|00\rangle + \sqrt{1-p}\,|01\rangle\,.$$

(a) Write down a representation of this mapping as a 4×4 matrix.

(b) Calculate the Kraus Operators $M_0 = \langle 0|U|0\rangle$ and $M_1 = \langle 1|U|0\rangle$ (here in our notation $|\alpha\beta\rangle$ we understand α to represent the system of interest and β to represent the environment).

[14 marks]

(ii) Prove the no-cloning theorem (i.e. that there is no Unitary map U that maps $|\psi\rangle\,|\phi\rangle$ to $|\psi\rangle\,|\psi\rangle$). **[11 marks]**

4. (i) Draw the quantum circuit corresponding to each of the following operations on the given input. (You do not need to calculate the resulting output.)

(a) $(H \otimes X)\texttt{CNOT}(1,2)(H \otimes H)\,|11\rangle$

(b) $(X \otimes Y \otimes H)\texttt{SWAP}(1,3)(CR_{\pi/7}(1,2))\,|000\rangle$

(c) $(Y^{\otimes 2})\texttt{SWAP}(Y \otimes I)\,|00\rangle$

[15 marks]

(ii) Calculate the permutation matrix U_f used in the Deutsch-Jozsa algorithm for the function $f : \{0,1,2,3,4,5,6,7\} \to \{0,1\}$, given by

$$f(0) = f(4) = f(6) = f(7) = 0$$
$$f(1) = f(2) = f(3) = f(5) = 1.$$

[10 marks]

5. Suppose that we want to find the letter 'y' in the (unordered) list L = [s, t, b, u, a, y, o, p].

(i) Using a simple classical algorithm - how many comparisons do we need to make in the

(a) best case scenario?

(b) worst case scenario?

(c) average case scenario?

[9 marks]

(ii) Using the Grover Algorithm, determine the probability of success (i.e. of finding the letter 'y') after (a) one operation, (b) two operations. **[16 marks]**